

Harrow Bay Offshore Wind – Energy Yield Assessment

P50 / P90 – financial close basis

Sofie Lindqvist · 05 December 2025 · HB-YLD-RPT-0004

Executive summary

The net P50 annual energy production of Harrow Bay Offshore Wind (40 × Halden HTS-12, 480 MW) is **1,900 GWh/a**, corresponding to a net capacity factor of 45.2 %. The 10-year P90 is **1,689 GWh/a** and the 1-year P90 is 1,641 GWh/a. These figures are the basis of the base case in the financial model at financial close and of the volume assumptions in the PPA with Arcturus Steel (WE-PPA-2026-004, 150 MW pay-as-produced, ~590 GWh/a).

Result	GWh/a	Capacity factor
Gross energy (free stream, no losses)	2,287	54.4 %
Net P50	1,900	45.2 %
Net P75 (10-year)	1,789	42.5 %
Net P90 (10-year)	1,689	40.2 %
Net P90 (1-year)	1,641	39.0 %

Table 1 – Energy yield results

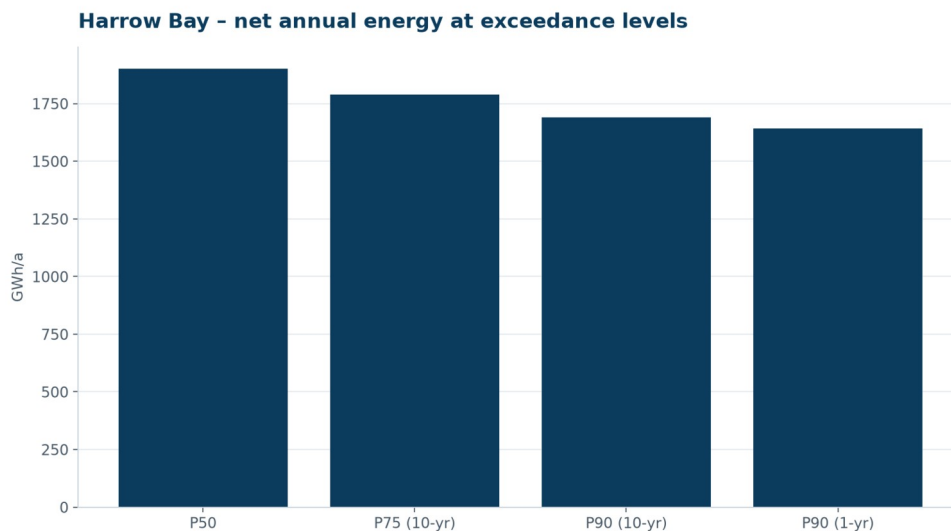


Figure 1 – Net energy at different exceedance levels

1. Wind farm configuration

- Turbines: 40 × Halden HTS-12 (12 MW), rotor diameter 236 m, hub height 140 m.
- Foundations: Monopiles, 9.5 m diameter, water depth 28–36 m.
- Layout: two staggered grids with 1.6 km (6.8D) spacing along the prevailing SW–W direction and 1.2 km across.
- Power curve: HTS-12 standard mode, air density 1.226 kg/m³, warranted power curve per WE-CON-2025-022.

2. Wind resource

Taken from HB-MET-RPT-0003 Rev 2: long-term mean wind speed 10.1 m/s at 140 m, Weibull A = 11.4 m/s, k = 2.18. Wind speed uncertainty of 3.4 % translates to roughly 4.5 % in energy (sensitivity ratio 1.3 at this site).

3. Wake modelling

Wake losses were calculated with the TurbOPark model, calibrated against SCADA data from three operating North Sea wind farms with similar spacing, and cross-checked with an engineering RANS model for the most frequent directions. Internal wake losses are 7.6 %; the external wake from the neighbouring Dogger cluster projects (existing and consented) adds 1.6 %, giving total **wake losses of 9.2 %**.

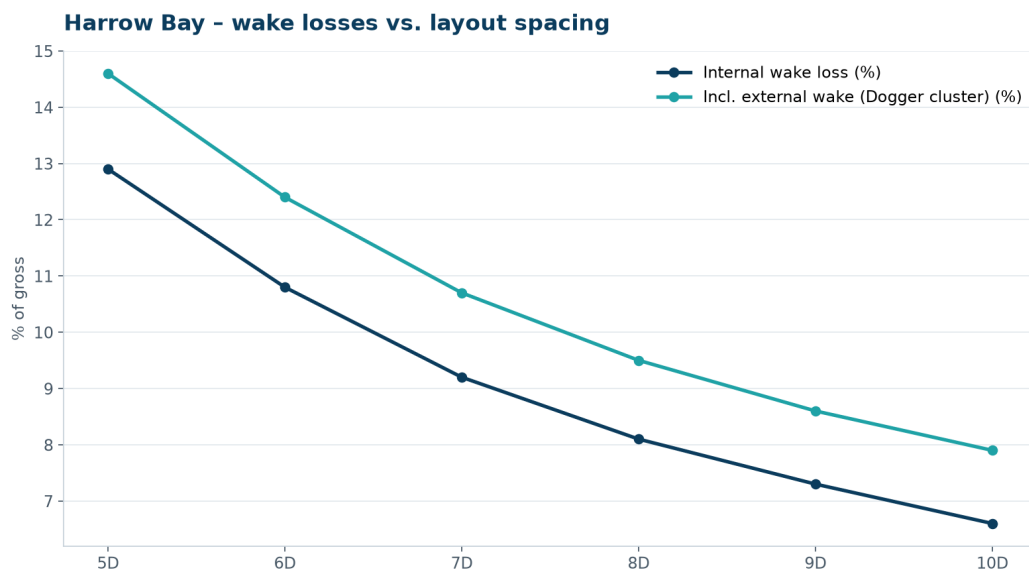


Figure 2 – Sensitivity of wake losses to turbine spacing (rotor diameters $D = 236\text{ m}$)

A denser layout (6D) would have saved about 4 km of 66 kV array cable but increased wake losses by 1.6 percentage points – roughly 30 GWh/a, far more than the cable saving. The chosen 6.8D spacing was confirmed in the layout freeze of September 2025.

4. Loss assumptions

Loss category	P50 assumption
Wake losses (internal and external)	9.2 %
Turbine availability	3.5 %
Electrical losses to grid meter	2.4 %
Turbine performance (power curve, hysteresis)	1.5 %
Environmental (icing, blade degradation, high-wind hysteresis)	0.8 %
Curtailement (grid and seabird curtailement)	0.6 %
Total losses	16.9 %

Table 2 – Loss assumptions (multiplicative)

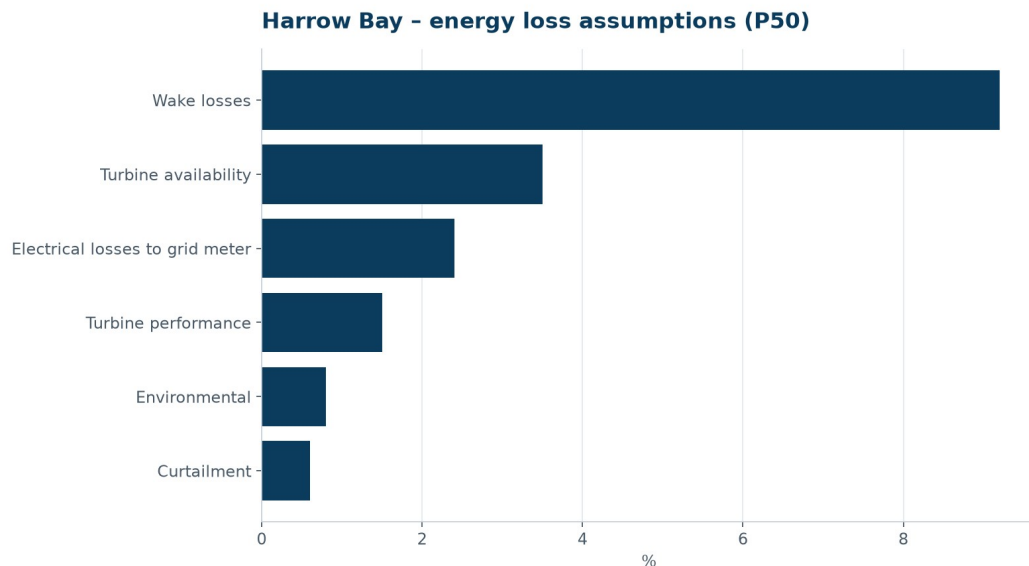


Figure 3 – Loss assumptions

Availability of 96.5 % reflects the warranted availability under the service agreement with Halden Turbine Systems plus balance-of-plant downtime (export cable faults modelled at one outage per cable per 10 years with 60 days repair time). Curtailment includes a small allowance for seabird-related curtailment during peak migration, as discussed with Marlow Ecology.

5. Uncertainty

Component	Energy uncertainty
Wind measurement (floating LiDAR)	4.5 %
Long-term correction (ERA5, 22 years)	3.2 %
Vertical and horizontal extrapolation	3.0 %
Wake and flow model	4.0 %
Loss assumptions	2.5 %
Power curve	3.0 %
Combined (excl. inter-annual variability)	8.4 %
Inter-annual variability – 1 year	6.5 %
Inter-annual variability – 10 years	2.1 %
Total – 1 year	10.6 %
Total – 10 years	8.7 %

6. Conclusions

- Net P50 1,900 GWh/a – enough for about 520,000 homes.
- Debt sizing at P90 (1-year) of 1,641 GWh/a, as agreed with Hanseatic Infrastructure Bank.
- The largest single lever on yield remains wake losses; the external wake should be re-assessed if the planned extension of the neighbouring cluster is consented.
- Recommendation: commission a post-construction yield reconciliation after 12 months of full operation.